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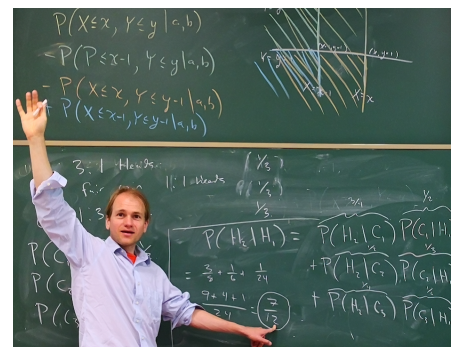
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Last update: 23rd June 2026

Professional Experience

Current Academic and Leadership Appointments

- 2024 – ♦ **Lead Biostatistician** for Pediatric Oncology Clinical Trials at Penn State
- 2024 – 2026 ♦ **Member** Penn State University Promotion and Tenure Review Committee
- 2022 – ♦ **Professor (with tenure)** *Pennsylvania State University*, College of Medicine, [Department of Public Health Sciences](#), Division of Biostatistics and Bioinformatics
- ♦ **Professor** *Department of Surgery*
- ♦ **Professor** *Department of Neurosurgery*
- ♦ **Professor** *Department of Family and Community Medicine*
- ♦ **Professor** *Department of Statistics*
- 2009 – ♦ **Member** *Penn State Cancer Institute*
- ♦ **Member** *Cancer Institute, Cancer Control Program*
- ♦ **Member** *Huck Institutes of the Life Sciences*
- ♦ **Member** *Bioinformatics and Genomics PhD Program Faculty*
- ♦ **Member** *Biomedical Sciences PhD Program Faculty*

Past Academic and Leadership Appointments

- 2024 – 2025 ♦ **Member** Penn State Cancer Institute Data Safety and Monitoring Committee
- 2011 – 2024 ♦ **Director** *Biostatistics PhD Program*, Penn State College of Medicine
- 2019, 2022 (Su) ♦ **Guest Instructor** 数理学院, 北京化工大学 (Beijing University of Chemical Technology)
- 2013 – 2022 ♦ **Associate Professor (with tenure)** *Pennsylvania State University*, Department of Public Health Sciences, Division of Biostatistics and Bioinformatics
- 2019 (Su) ♦ **Faculty Instructor** *MathPath*, Grand Valley State University
- 2011 – 2015 ♦ **Leader** *Biomedical Informatics Program*, Penn State Clinical and Translational Science Institute
- 2011 – 2015 ♦ **Director** *Bioinformatics Core*, Penn State College of Medicine
- 2011 (Su) ♦ **Visiting Scholar** 统计科学中心, 北京大学 (Peking University)
- 2009 – 2013 ♦ **Assistant Professor (tenure-track)** *Pennsylvania State University*, Department of Public Health Sciences, Division of Biostatistics and Bioinformatics
- 2007 – 2009 ♦ **Assistant Professor (tenure-track)** *University of Florida*, Department of Statistics
- 2007 (Wi) ♦ **Associate Instructor** *University of California, San Diego*, Department of Mathematics
- 2005 – 2007 ♦ **Adjunct Instructor** *San Diego Miramar College*, Department of Mathematics
- 2002 – 2007 ♦ **Teaching Assistant** *University of California, San Diego*, Department of Mathematics
- 2001 – 2003 (Su) ♦ **Counselor** *Program in Mathematics for Young Scientists*, Boston University
- 1999 – 2002 ♦ **Teaching Assistant** *University of Southern California*, Department of Mathematics

Professional Experience (continued)

Current Editorial Board Positions

- 2024 – ◇ Editorial Board Member for *Statistics Innovation*
- 2024 – ◇ Editorial Board Member for *International Chinese Language Education Communications* 《国际中文教育通讯》

Music

- 2025/09 – 2026/08 ◇ **Director of Music** *Trinity Evangelical Lutheran Church* in Lemoyne, PA
- 2025/01 – 2025/09 ◇ **Organist & Pianist** *St. Peter's Lutheran Church* in Highspire, PA
- 2025/01 – 2025/05 ◇ **Organist & Pianist** *Salem Lutheran Church* in Oberlin, PA
- 2020 – 2024 ◇ **Organist & Pianist** *Renewed in Grace Co-Op* (Trinity Lutheran Church in Steelton, PA & St. Peter's Lutheran Church in Highspire, PA)
- 2018 – 2020 ◇ **Choir Director, Organist & Pianist** *Trinity Lutheran Church* in Steelton, PA
- 2015 – 2018 ◇ **Pianist** (every other week) *Paxton Church of the Brethren* in Harrisburg, PA
- 1994 – 1998 ◇ **Pianist** (early services) *Our Saviour's Lutheran Church* in Long Beach, CA

Other Professional Services

- ◇ **AI-Powered Research & Workflow Consulting** Work with researchers, professionals, and organizations across a wide range of fields to bring modern generative-AI tooling into their work in a thoughtful, practical way — using both mainstream cloud-based AI services and locally-hosted large-language-model deployments depending on what fits the work.
- ◇ **Expert Witness Services** Biostatistical expert opinions and testimony in medical malpractice, pharmaceutical, and scientific causation matters — written expert reports, deposition and trial testimony, Daubert/Frye admissibility analyses, and case consultation with counsel on statistical strategy.
- ◇ **Independent Data Monitoring** Have served as an independent biostatistician on Data Safety and Monitoring Committees (DSMCs) and Data Monitoring Committees (DMCs) for investigator-initiated and industry-sponsored clinical trials, including pre-trial review of monitoring plans and stopping rules, unblinded interim analyses, and statistical reports to DSMC/DMC deliberations.

Education

- 2002 – 2007 ◇ **Ph.D., Mathematics** *University of California, San Diego*
Dissertation: *Nonparametric Function Estimation with Infinite-Order Kernels and Applications*
Advisor: Dimitris N. Politis
- 2002 – 2005 ◇ **M.S., Statistics** *University of California, San Diego*
- 1998 – 2002 ◇ **M.A., Mathematics** *University of Southern California*
- 1998 – 2002 ◇ **B.S., Mathematics** *University of Southern California*
- 1996 – 1998 ◇ *North Carolina School of Science and Mathematics* (residential high school)

Publications

- 195 **Berg, A.** (2025). Entropy-based sound-character mapping for Chinese character learning. *International Journal of Chinese Language Teaching*, 6: (2), 63–89.
 10.46451/ijclt.20250205.
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- 194 Brenman, J. S., **Berg, A.**, Heinemann, A. W., Chen, Y., Felix, E. R., & Farkas, G. J. (2025). Associations between obesity and cardiometabolic, physical, and psychosocial health in chronic traumatic spinal cord injury. *Archives of Physical Medicine and Rehabilitation*.
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- 193 Donato, Z., Levy, A., Sneij, A., LaVela, S., **Berg, A.**, Shapiro, L., & Farkas, G. (2025). Grocery shopping behavior and eating behavior differ between persons with and without a spinal cord injury. *Topics in Spinal Cord Injury Rehabilitation*, **31**: (1), 66–76.
[10.46292/sci24-00028](https://doi.org/10.46292/sci24-00028).
- 192 Farkas, G. J., Caldera, L. J., Hodgkiss, D. D., Mitchell, J. R., Pelaez, T. F., Cusnier, M. A., Cole, A. J., Daniel, S. G., Farrow, M. T., Gee, C. M., Kincaid-Sharp, E. A., Logan, A. M. G., McMillan, D. W., Nightingale, T. E., Perdue, B., Portes, P., Walson, F. T., Volmrich, A. M., Reynolds, J. M., ... **Berg, A.** (2025). Cardiometabolic risk in chronic spinal cord injury: A systematic review with meta-analysis and temporal and geographical trends. *Journal of clinical medicine*, **14**: (2872), 1–31.
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- 191 Farkas, G. J., Cunningham, P. M., **Berg, A.**, Jimshelishvili, G., Mendez, A. J., Gater, D. R., Nash, M. S., & Rolls, B. J. (2025). Gastrointestinal dysmotility and impaired gut peptide-satiety coupling in men with spinal cord injury. *Physiology & Behavior*, **305**: 115207.
[10.1016/j.physbeh.2025.115207](https://doi.org/10.1016/j.physbeh.2025.115207).
- 190 Gill, S., Ho, D., Veldheer, S., **Berg, A.**, Morgis, R., Curry, W., & Rabago, D. (2025). Barriers and facilitators to topical fluoride varnish application in well-child visits. *JDR Clinical & Translational Research*, **10**: (1 Suppl), 46S–53S.
[10.1177/23800844251328646](https://doi.org/10.1177/23800844251328646).
- 189 Lato, A. T., **Berg, A.**, Parascando, J. A., Phillips, S. F., Onks, C. A., & Loeffert, J. R. (2025). Enhancing knowledge and confidence in wilderness medicine through virtual education: Lessons from a national boot camp. *Cureus*, **17**: (2).
[10.7759/cureus.78689](https://doi.org/10.7759/cureus.78689).
- 188 Yee, N. S., **Berg, A.**, Benson, A. B., Turk, A., Boland, P. M., & Peterson, C. (2025). A phase ii study of liposomal irinotecan in combination with oxaliplatin, leucovorin, and 5-fluorouracil (nalirifox) in locally advanced pancreatic carcinoma: Btrcr-gi15-o67. *JCO Oncology Advances*.
[10.1200/OA-25-00047](https://doi.org/10.1200/OA-25-00047).
- 187 Allen, E., Goldstein, E., Lennon, R., **Berg, A.**, Regner, C., Eiden, R., Symons Downs, D., Neiderhiser, J., & Zgierska, A. (2024). COVID-19 pandemic-related perceived loneliness as a potential risk factor for worse outcomes among people who are pregnant or postpartum. *Wisconsin Medical Journal*, **123**: (6), 515–520.
- 186 **Berg, A.** (2024a). Causal inference from high-dimensional static data in soil microbiota networks: Comment on "topological change of soil microbiota networks for forest resilience under global warming" by gong et al. *Physics of life reviews*, **51**: 281–282.
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- 185 **Berg, A.** (2024b). Quantitative methods for precision medicine: Pharmacogenomics in action: Rongling wu, boca raton, fl: Chapman & Hall/CRC Press [Book review]. *Journal of the American Statistical Association*, **119**: (546), 1690–1691.
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- 184 Farkas, G. J., **Berg, A.**, Sneij, A., Dolbow, D. R., Gorgey, A. S., & Gater, D. R. (2024). The comparison of total energy and protein intake relative to estimated requirements in chronic spinal cord injury. *British Journal of Nutrition*, **131**: (3), 489–499.
[10.1017/S0007114523002088](https://doi.org/10.1017/S0007114523002088).
- 183 Hawila, N., & **Berg, A.** (2024). Bayesian joint modeling of interrater and intrarater reliability with multilevel data. *Asian Journal of Statistics and Applications*, **1**: (1), 85–99.
<https://www.pbjournals.com/ajsa/issue/89>.
[Hawila, Berg_2024_Asian-Journal-of-Statistics-and-Applications.pdf](#).
- 182 Healy, M. E., Messer, L. H., Stasinopoulos, M. G., Brewer, S. E., **Berg, A.**, & Oser, T. K. (2024). Understanding patient and health care provider perceptions of barriers and facilitators to diabetes care in immigrant and refugee populations. *Discover Public Health*, **21**: (88), 1–12.
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- 181 McGuire, D., Markus, H., Yang, L., Xu, J., Montgomery, A., **Berg, A.**, Li, Q., Carrel, L., Liu, D. J., & Jiang, B. (2024). Dissecting heritability, environmental risk, and air pollution causal effects using >50 million individuals in MarketScan. *Nature Communications*, **15**: (1), 5357.
[10.1038/s41467-024-49566-6](https://doi.org/10.1038/s41467-024-49566-6).
- 180 Onks, C., Weaver, L., Latorre, J., Silvis, M., **Berg, A.**, Phillips, S., Loeffert, J., French, C., & Armstrong, A. (2024). The most effective corticosteroid dose in the treatment of glenohumeral osteoarthritis: Feasibility pilot and protocol for double blinded randomized controlled trial. *Osteoarthritis and Cartilage Open*, **6**: (3), 100484.
[10.1016/j.ocarto.2024.100484](https://doi.org/10.1016/j.ocarto.2024.100484).
- 179 Vangala, V., Chen, Y.-C., Dinavahi, S. S., Gowda, K., Lone, N. A., Herlyn, M., Drabick, J., Helm, K., Zhu, J., Neves, R. I., et al. (2024). Tumor heterogeneity shapes survival dynamics in drug-treated cells, revealing size-drifting subpopulations. *ACS Pharmacology & Translational Science*, **7**: (11), 3573–3584.
[10.1021/acsptsci.4c00453](https://doi.org/10.1021/acsptsci.4c00453).
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- 177 Farkas, G. J., Cunningham, P. M., Sneij, A. M., Hayes, J. E., Nash, M. S., **Berg, A.**, Gater, D. R., & Rolls, B. J. (2023). Reasons for meal termination, eating frequency, and typical meal context differ between persons with and without a spinal cord injury. *Appetite*, **192**: (107110), 1–9.
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- 175 Karpa, K., Ward, J., Stegman, M., **Berg, A.**, & Leong, S. L. (2023). Ipex: A gamification tool for learner application of pharmacologic principles of opioid use, misuse, and addiction. *Pharmacology Research & Perspectives*, **11**: (5), e01141.
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- 171 Walser, S. A., Costigan, H., Stuckey, H. L., **Berg, A.**, & Stephens, M. B. (2023). The opaque language of sexuality: Medical students' and providers' beliefs about virginity. *Archives of Sexual Behavior*, **52**: 2619–2638.
[10.1007/s10508-023-02578-7](https://doi.org/10.1007/s10508-023-02578-7).
- 170 Wang, H., Berger, K. N., Miller, E. L., Fu, W., Broglie, L., Goldman, F. D., Konig, H., Lim, S. J., **Berg, A.**, Talano, J.-A., Comito, M. A., Farag, S. S., & Pu, J. J. (2023). The impacts of total body irradiation on umbilical cord blood hematopoietic stem cell transplantation. *Therapeutic Advances in Hematology*, **14**: 1–13.
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- 168 Dinavahi, S. S., Chen, Y.-C., Punmath, K., **Berg, A.**, Herlyn, M., Foroutan, M., Huntington, N. D., & Robertson, G. P. (2022). Targeting WEE1/AKT restores p53-dependent NK cell activation to induce immune checkpoint blockade responses in 'cold' melanoma. *Cancer Immunology Research*, **10**: (6), 757–769.
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- 167 Dinavahi, S. S., Chen, Y.-C., Gowda, R., Dhanyamraju, P. K., Punmath, K., Desai, D., **Berg, A.**, Kimball, S. R., Amin, S., Yang, J., & Robertson, G. P. (2022). Targeting protein translation in melanoma by inhibiting EEF-2 kinase regulates cholesterol metabolism through SREBP2 to inhibit tumor development. *International Journal of Molecular Sciences*, **23**, 3481: (1-19).
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[10.22454/FamMed.2022.324428](https://doi.org/10.22454/FamMed.2022.324428).
- 162 Schlegel, D., Heron, C., **Berg, A.**, Parascando, J., & Silvis, M. (2022). Diagnoses per encounter by telephone, televideo, and in-office visits. *Journal of the American Board of Family Medicine*, **35**: (3), 491–496.
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[10.1097/dcr.0000000000002273](https://doi.org/10.1097/dcr.0000000000002273).
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- 159 **Berg, A.** (2021b). Bayesian modeling competitions for the classroom. *Colombian Journal of Statistics*, **44**: (2), 243–252.
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- 157 **Berg, A.**, & Hawila, N. (2021a). Introducing Bayesian inference with the taxicab problem (A. Bilgin & H. McGillivray, Eds.). *Proceedings of the 10th Australian Conference on Teaching Statistics*, 55–60.
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- 156 **Berg, A.**, & Hawila, N. (2021b). Some teaching resources using R with illustrative examples exploring COVID-19 data. *Teaching Statistics*, **43**: S98–S109.
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- 147 Liao, J., **Berg, A.**, & McMurry, T. L. (2021). A robustified posterior for Bayesian inference on a large number of parallel effects. *The American Statistician*, **75**: (2), 145–151.
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Extramural Support

Current Support

- 2025 – 2027 ♦ Biostatistician, **BCC022: Phase II Trial of Naxitamab and Tipifarnib for Relapsed/Refractory Neuroblastoma** (Cannonball Kids' Cancer Foundation) Total Budget: \$200,000. Award #: 293814. PI: Giselle Saulnier Sholler.

Extramural Support (continued)

- 2025 – 2027 ◇ Biostatistician, **BCCo22: Phase II Trial of Naxitamab and Tipifarnib for Relapsed/Refractory Neuroblastoma — Trial Blazers portion** (Trial Blazers for Kids Foundation) Total Budget: \$133,300. Award #: 298322. PI: Giselle Saulnier Sholler.
- 2024 – 2027 ◇ Biostatistician, **A Unique Small Molecule for Triple-Negative Breast Cancer** (U.S. Army Medical Research and Development Command) Total Budget: \$1,255,505. Award #: HT94252410698. PI: Arun Sharma.
- 2023 – 2026 ◇ Biostatistician, **Identifying Novel Approaches for Melanoma Prevention Using Mouse Models with Humanized Telomeres** (U.S. Army Medical Research and Development Command) Total Budget: \$829,986. Award #: HT94252310964. PI: Gavin Robertson.
- 2023 – 2026 ◇ Biostatistician, **Combination Therapy for Childhood Acute Myeloid Leukemia** (Four Diamonds Fund Research Program) Total Budget: \$250,000. Award #: 4D2I_2024_1002-EXT. PI: Arati Sharma.
- 2021 – 2027 ◇ Co-Investigator, **Linking olfactory deficits to memory impairment and AD neurodegeneration** (National Institute on Aging) Total Budget: \$3,855,438. Award #: 5 RoI AG070088-05-EXT. PI: Prasanna Karunanayaka.

Pending Grants

- 12/2026 – 11/2031 ◇ Principal Investigator (Penn State subaward), **Nutrition and cardiometabolic health after spinal cord injury** (University of Miami) Total Budget: \$247,984.
- 12/2026 – 11/2031 ◇ Biostatistician, **Cellular-plasticity mechanisms in cancer migration** (National Institutes of Health) Total Budget: \$7,312,233. PI: Gavin Robertson.
- 12/2026 – 11/2031 ◇ Biostatistician, **Training program in translational pediatric oncology** (National Institutes of Health) Total Budget: \$1,045,142. PI: Wei Li.
- 12/2026 – 11/2031 ◇ Principal Investigator (Penn State subaward), **Nanotherapeutics for drug-resistant and metastatic osteosarcoma** (Brandeis University) Total Budget: \$125,123.
- 10/2026 – 09/2029 ◇ Biostatistician, **Persister-cell targeting in melanoma metastasis** (U.S. Army Medical Research and Development Command) Total Budget: \$1,263,649. PI: Gavin Robertson.
- 10/2026 – 09/2028 ◇ Biostatistician, **Immuno-oncology drug combinations** (American Cancer Society) Total Budget: \$297,000. PI: Gavin Robertson.
- 10/2026 – 09/2028 ◇ Biostatistician, **Parental decision-making in pediatric oncology trials** (American Cancer Society) Total Budget: \$297,000. PI: Ariane Levesque.
- 08/2026 – 07/2027 ◇ Biostatistician, **Patient–provider communication in pediatric oncology** (The Andrew McDonough B+ Foundation) Total Budget: \$25,000. PI: Ariane Levesque.
- 07/2026 – 06/2031 ◇ Biostatistician, **Pharmacological intervention for major depressive disorder** (National Institutes of Health) Total Budget: \$6,715,017. PI: Dahlia Mukherjee.
- 07/2026 – 06/2031 ◇ Principal Investigator (Penn State subaward), **Metabolic therapeutic vulnerabilities in pediatric sarcoma** (Albert Einstein College of Medicine) Total Budget: \$123,822.
- 07/2026 – 06/2031 ◇ Biostatistician, **Autophagy and intestinal disease mechanisms** (National Institutes of Health) Total Budget: \$3,884,722. PI: Prashant Nighot.
- 07/2026 – 06/2031 ◇ Biostatistician, **Immune-modulatory strategies in metastatic melanoma** (National Institutes of Health) Total Budget: \$3,914,975. PI: Gavin Robertson.
- 07/2026 – 06/2031 ◇ Biostatistician, **Melanoma progression biology** (National Institutes of Health) Total Budget: \$3,946,982. PI: Gavin Robertson.
- 07/2026 – 06/2031 ◇ Biostatistician, **Targeted inhibitors for melanoma** (National Institutes of Health) Total Budget: \$3,914,950. PI: Arun Sharma.
- 06/2026 – 05/2028 ◇ Biostatistician, **Pediatric melanoma modeling** (Melanoma Research Foundation) Total Budget: \$250,000. PI: Gavin Robertson.
- 04/2026 – 03/2031 ◇ Principal Investigator (Penn State subaward), **Nanotherapeutic delivery for pediatric bone cancer** (Brandeis University) Total Budget: \$123,630.
- 04/2026 – 03/2031 ◇ Biostatistician, **Drug-delivery strategies for pediatric leukemia** (National Institutes of Health) Total Budget: \$3,946,935. PI: Arati Sharma.

Extramural Support (continued)

01/2026 – 12/2026	◇ Biostatistician, Immune targeting of drug-tolerant cancer cell populations (Elsa U. Pardee Foundation) Total Budget: \$205,589. PI: Gavin Robertson.
12/2025 – 11/2030	◇ Biostatistician, Thermal physiology and aging (National Institutes of Health) Total Budget: \$3,811,428. PI: William Kenney.
12/2025 – 11/2027	◇ Biostatistician, Targeting drug-tolerant cancer cell populations (National Institutes of Health) Total Budget: \$458,704. PI: Gavin Robertson.
10/2025 – 10/2027	◇ Biostatistician, Biomarker development for melanoma (Melanoma Research Foundation) Total Budget: \$250,000. PI: Gavin Robertson.
09/2025 – 08/2030	◇ Principal Investigator (Penn State subaward), Metabolic therapeutic vulnerabilities in pediatric sarcoma (Albert Einstein College of Medicine) Total Budget: \$91,269.
09/2025 – 08/2028	◇ Principal Investigator (Penn State subaward), Dietary response and obesity in pediatric disease (University of Miami) Total Budget: \$77,249.

Completed Grants

2020 – 2026	◇ Biostatistician, Targeting Aldehyde Dehydrogenase for Cancer Prevention (National Cancer Institute) Total Budget: \$3,170,470. Award #: 5 Ro1 CA241148-05. PI: Gavin Robertson.
2020 – 2025	◇ Biostatistician, A Phase II Study to Evaluate the Response and Tolerability of Abemaciclib in Patients with Advanced Biliary Tract Carcinoma Who Have Failed Prior Chemotherapy (Lilly USA LLC) Total Budget: \$383,500. Award #: I3Y-MC-E004. PI: Nelson Yee.
2022 – 2025	◇ Biostatistician, Neuro-pharmacological Properties of Repurposed Posaconazole in Glioblastoma: A Phase o Clinical Trial (National Cancer Institute) Total Budget: \$140,846. Award #: 5 Ro3 CA255992-02-EXT. PI: Seyed Mansouri.
2023	◇ Biostatistician, Characterization of Passive and Active Whole-Body Heat Stress Responses in Obese and Non-Obese Adults (National Institute on Aging) Total Budget: \$215,064. Award #: 1 F32 AG079581-01A1. PI: William Kenney.
2022 – 2023	◇ Biostatistician, RAD-PD: Registry for the Advancement of DBS in Parkinson's Disease (Icahn School of Medicine at Mount Sinai) Total Budget: \$29,281. Award #: 0266-3886-4609. PI: James McInerney.
2020 – 2022	◇ Co-Investigator, Primary Care Training and Enhancement (Health Resources and Services Administration) Total Budget: \$2,568,830. Award #: 5 ToBHP30010-05-00-EXT. PI: Shou Ling Leong.
2019 – 2022	◇ Biostatistician, A Novel Agent for Lung Cancer Prevention (U.S. Army Medical Research and Development Command) Total Budget: \$582,205. Award #: W81XWH1910231-EXT. PI: Arun Sharma.
2018 – 2021	◇ Biostatistician, A Promising Small Molecule for Pancreatic Cancer Therapy (National Cancer Institute) Total Budget: \$362,705. Award #: 1 R21 CA234681-01. PI: Arun Sharma.
2019 – 2021	◇ Biostatistician, Modulating p53 Transcriptional Activity to Reduce Melanoma Brain Metastasis (Melanoma Research Foundation) Total Budget: \$100,000. Award #: 226521. PI: Saketh Dinavahi.
2019 – 2020	◇ Biostatistician, Establishing Drug-like Properties of AS-10, a Potential Pancreatic Cancer Therapy (Penn State Research Foundation) Total Budget: \$12,500. Award #: 224862. PI: Arun Sharma.
2019 – 2020	◇ Co-Investigator, Primary Care Training and Enhancement Program (Health Resources and Services Administration) Total Budget: \$975,578. Award #: 5 ToBHP30010-04-00. PI: Shou Ling Leong.
2017 – 2019	◇ Co-Investigator, Area Health Education Centers (State Baseline) (PA Department of Health) Total Budget: \$1,230,000. Award #: 4100080688. PI: Alan Adelman.
2018 – 2019	◇ Co-Investigator, Primary Care Training and Enhancement Program (Health Resources and Services Administration) Total Budget: \$477,871. Award #: 5 ToBHP30010-03-00. PI: Shou Ling Leong.
2017 – 2019	◇ Co-Investigator, Spinal Cord Injury Exercise and Nutrition Conceptual Engagement (SCIENCE) (National Institute of Child Health and Human Development) Total Budget: \$1,982,948. Award #: 1 Ro1 HD091278-01-REVISED. PI: David Gater.
2018	◇ Biostatistician, Nucleus Accumbens-Associated Protein-1 in Melanoma Immunotherapy (National Cancer Institute) Total Budget: \$2,835,828. Award #: 1 Ro1 CA221867-01A1. PI: Jin-Ming Yang.

Extramural Support (continued)

- 2013 – 2018 ◇ Biostatistician, [Macronutrient Regulation of Alternative Pre-mRNA Splicing](#) (National Institute of Diabetes and Digestive and Kidney Diseases) Total Budget: \$330,546. Award #: 1 Ro1 DK094141-01A1. PI: Scot Kimball.
- 2016 – 2018 ◇ Co-Investigator, [Stem Cell-Derived Regulatory T Cells for Therapeutic Use in Arthritis](#) (National Institute of Allergy and Infectious Diseases) Total Budget: \$1,915,070. Award #: 1 Ro1 AI121180-01A1. PI: Jianxun Song.
- 2014 – 2018 ◇ Biostatistician, [A Diagnostic Plasma Protein Panel for Alcohol Abuse](#) (National Institute on Alcohol Abuse and Alcoholism) Total Budget: \$344,250. Award #: 1 Ro1 AA021661-01A1. PI: Kent Vrana.
- 2015 – 2017 ◇ Biostatistician, [Improving Autism Screening with Brain-Related miRNA](#) (Motion Intelligence LLC) Total Budget: \$18,000. Award #: 181047 EXT. PI: Steven Hicks.
- 2015 – 2017 ◇ Biostatistician, [A Novel Compound for Colorectal Cancer Prevention](#) (National Cancer Institute) Total Budget: \$74,443. Award #: 5 Ro3 CA181845-02. PI: Arun Sharma.
- 2016 ◇ Biostatistician, [Stem Cell-Derived Regulatory T Cells for Therapeutic Use of Autoimmune Diabetes](#) (American Diabetes Association) Total Budget: \$115,000. Award #: 1-16-IBS-281. PI: Jianxun Song.
- 2012 – 2016 ◇ Co-Investigator, [Relevance of PIG-A Mutations in Acquired Bone Marrow Failure Diseases](#) (Aplastic Anemia and MDS International Foundation) Total Budget: \$60,000. Award #: 146818. PI: Jeffrey Pu.
- 2014 – 2016 ◇ Biostatistician, [Chemoprevention of Colon Cancer by a Novel COX-2 Inhibitor](#) (National Cancer Institute) Total Budget: \$76,500. Award #: 1 Ro3 CA180539-01A1. PI: Dhimant Desai.
- 2013 – 2015 ◇ Co-Investigator, [Development of Bioinformatics Methods for Medical Research](#) (PA Tobacco Settlement Fund) Total Budget: \$4,551,311. Award #: 4100057673. PI: Arthur Berg.
- 2011 – 2015 ◇ Co-PI, [Penn State Clinical and Translational Science Institute](#) (National Institutes of Health) Total Budget: \$4,467,207. Award #: 1 UL1 RR033184-01. PI: Lawrence Sinoway.
- 2012 – 2014 ◇ Co-Investigator, [Development of a Novel Agent for Lung Cancer Prevention](#) (National Cancer Institute) Total Budget: \$71,053. Award #: 1 Ro3 CA171646-01. PI: Arun Sharma.
- 2011 – 2012 ◇ Co-PI, [Using High Dimensional Models to Predict 30-Day Readmission after Heart Failure](#) (HMC Clinical Research Incentive Grant) Total Budget: \$49,900. Award #: 133476. PI: Robert Aronoff.
- 2010 – 2011 ◇ Co-Investigator, [Correlation of Gene Haplotype with Complications after IPAA](#) (American Society of Colon and Rectal Surgeons) Total Budget: \$50,000. Award #: 1-104. PI: Rishabh Sehgal.

Presentations

- 2025 ◇ [Entropy-based Sound-Character Mapping for Chinese Character Learning Through Phonological Analysis](#) presentation at the [Chinese Language Teachers Association \(CLTA\) Annual Conference](#) in New York City, May 9-10, 2025.
- 2025 ◇ [基于熵值分析的汉语音形映射：通过语音分析提升语言学习](#) presentation at the [8th International Forum on Linguistics and Chinese Education \(IFOLCE\)](#) / 32nd International Conference on Chinese Language Instruction at Princeton University, April 5-6, 2025.
- 2025 ◇ [Targepeutics DIPG Trial—Phase II Primary Endpoint](#) presentation at the [Beat Childhood Cancer Annual Meeting](#) held in St. Petersburg, March 5-7, 2025.
- 2024 ◇ [End of Induction Response Correlates with Clinical Outcomes In Patients with High-Risk Neuroblastoma](#) oral presentation at the [Beat Childhood Cancer Annual Meeting](#) in February 2024.
- 2021 ◇ [Statistical Analysis of the International Mathematical Olympiad](#) oral presentation at the [Mathematical Association of America MathFest Conference](#) in August 2021.
- 2021 ◇ [Introducing Bayesian Inference with the Taxicab Problem](#) oral presentation at the [10th Australian Conference on Teaching Statistics \(OZCOTS 2021\)](#) in July 2021.
- 2021 ◇ [Sharpening Jensen’s Inequality with Applications](#) oral presentation at the [Australian Statistical Society and New Zealand Statistical Association Conference \(ANZSC 2021\)](#) in July 2021.
- 2020 ◇ [Describing the Biostatistics Career](#). Zoom presentation to students at [Hershey Middle School](#) for their career day event on April 28, 2020.

Presentations (continued)

- 2019 ◇ **Probability, Statistics, and Bayesian Analysis.** Summer short course presented to over 50 undergraduate students at [Beijing University of Chemical Technology](#) from July 1, 2019 – July 10, 2019.
- 2019 ◇ **Introduction to R and Monte Carlo Simulation.** Short course presented to very talented middle school students at the 2019 MathPath Summer Camp on the [Grand Valley State University](#) campus. July 21-28, 2019.
- 2019 ◇ **AIME Problem Solving.** Short course presented to very talented middle school students at the 2019 MathPath Summer Camp on the [Grand Valley State University](#) campus. July 21-28, 2019.
- 2019 ◇ **Advancing The Bayesian New Statistics Paradigm with Interval Null Hypotheses: Moving from p-values to Posterior Probabilities and Bayes Factors.** Poster presentation at [2019 United States Conference on Teaching Statistics](#) held at [Penn State](#) from May 16-18, 2019.
- 2018 ◇ **A new look at Bayes factor and its implication on model selection.** [Penn State Division of Biostatistics Seminar](#) on September 13, 2018. I worked with Dr. Jason Liao and PhD student Vishal Midya on this presentation but the seminar was led by Dr. Jason Liao.
- 2017 ◇ **Reduced Bias in Nonparametric Censored Density and Hazard Estimation.** 2017 Conference on Lifetime Data Science: Data Science, Precision Medicine and Risk Analysis with Lifetime Data held at the [University of Connecticut](#). <http://merlot.stat.uconn.edu/lida/>. May 25-27, 2017.
- 2017 ◇ **Robustified Bayesian Inference.** Penn State Division of Biostatistics Seminar at [Penn State Hershey](#) on April 27, 2017.
- 2016 ◇ **Sleep disordered breathing is associated with DNA methylation of obesity-related genes in adolescents.** Presentation at the European Sleep Research Annual Conference in [Bologna, Italy](#) from September 13 – 16, 2016.
- 2016 ◇ **A Statistical Potpourri: Swarthmorean Proportions, Fourier Transforms, Group Representations, and More.** Invited talk presented at [Swarthmore College](#), March 22, 2016.
- 2016 ◇ **DNA Methylation and Obesity and Sleep Apnea.** Penn State Division of Epidemiology Seminar at [Penn State Hershey](#) on September 21, 2016.
- 2016 ◇ **Bayesian approaches to clinical trial design.** Penn State Neurosurgery Department Evidence-Based Medicine Seminar at [Penn State Hershey](#) on March 18, 2016.
- 2015 ◇ **Superkernels, Symmetry, and Survival.** Penn State Division of Biostatistics & Bioinformatics Seminar at [Penn State Hershey](#) on October 15, 2015.
- 2014 ◇ **Design and Analysis of High-Throughput Genomic Data with Emphasis on DNA Methylation.** Presentation at the School of Public Health in [Shanghai Jiao Tong University](#), Shanghai, China from June 28 – 29, 2014.
- 2013 ◇ **The Brainchild of Bayes and Fisher: Empirical Bayes Methodology.** Penn State Biostatistics & Bioinformatics Divisional Seminar at [Penn State Hershey](#) on November 20, 2013.
- 2013 ◇ **Medical Genetics.** Introduction to Patient Oriented Research class presentation at [Penn State Hershey](#). September 13, 2013.
- 2013 ◇ **Biostatistics & Bioinformatics Research at Penn State.** Presentation for the MD/PhD students at [Penn State Hershey](#). January 17, 2013.
- 2012 ◇ **Medical Genetics.** Introduction to Patient Oriented Research class presentation at [Penn State Hershey](#). September 14, 2012.
- 2012 ◇ **Biostatistics & Bioinformatics at PSU College of Medicine.** Public Health Sciences graduate student presentation at [Penn State Hershey](#). August 23, 2012.
- 2012 ◇ **Biomedical Informatics Program Update.** Presentation to the CTSI External Advisory Board at [Penn State Hershey](#). April 17, 2012.
- 2012 ◇ **Reproducible Research for Biostatisticians.** Public Health Sciences Seminar – Biostatistics & Bioinformatics Division at [Penn State Hershey](#). January 19, 2012.
- 2012 ◇ **Empirical Bayes Inference Made Robust: picking up where James-Stein left off.** Invited talk presented at [Temple University](#), September 7, 2012.
- 2011 ◇ **Standing Between a Bayesian and a Frequentist: An Empirical Bayes Exploration of Movies, Baseball, and Long Beach Basketball.** Invited talk presented at [California State University, Long Beach](#) (Long Beach, CA), March 2011.
- 2011 ◇ **Standing Between a Bayesian and a Frequentist: An Empirical Bayes Exploration of Movies, Baseball, and Williams College.** Invited talk presented at [Williams College](#) (Williamstown, MA), January 2011.

Presentations (continued)

- 2011 ◇ **Medical Genetics.** Introduction to Patient Oriented Research class presentation at [Penn State Hershey](#). December 9, 2011.
- 2011 ◇ **The Design and Analysis of Laboratory-Based Investigations.** Neurosurgery Journal Club presentation at [Penn State Hershey](#). March 25, 2011.
- 2010 ◇ **New Functional Clustering Methodology and Applications.** Invited talk presented in the Statistics Department at [Peking University](#), Beijing, China, July, 2010.
- 2010 ◇ **Multiple Estimation Bias Adjustment with Application to GWAS.** Invited presenter at the Symposium of the Center for Computational Biology, [Beijing Forestry University](#), Beijing, China, July, 19, 2010.
- 2010 ◇ **A Bootstrap Test for Time Series Linearity.** Invited talk at Conference on Resampling Methods and High Dimensional Data in [College Station, TX](#) March 24 – March 25, 2010.
- 2010 ◇ **Multiple Estimation Bias Adjustment with Application to GWAS.** Invited talk presented at the [University of Arizona](#) (Tucson, AZ), October 2010.
- 2010 ◇ **Variety's the very spice of life, That gives it all its flavor.** Invited talk presented at Student Colloquium Series in the Mathematics Department at [Bucknell University](#), Lewisburg, PA, March 2010.
- 2010 ◇ **The Art and Craft of Biostatistics in the Berg Lab.** Invited presenter at the Penn State Cancer Institute Molecular Epidemiology Working Group Meeting at [Penn State Hershey](#), November 15, 2010.
- 2010 ◇ **Biostatistical collaborations with COM researchers.** Presentation for College of Medicine incoming graduate students at [Penn State Hershey](#), August 16, 2010.
- 2010 ◇ **Statistical and Computational Pharmacogenomics.** Presentation at the Center for Pharmacogenetics Workshop, [Pennsylvania State University](#), University Park, PA, March 2010.
- 2010 ◇ **Gene Expression Functional Clustering and Expression QTLs.** Invited talk presented at the Statistics Department Seminar Series at [Pennsylvania State University](#), University Park, PA, February 2010.
- 2009 ◇ **Improving the Kaplan-Meier Estimator.** Invited talk at the 2009 AMS Sectional Meeting on Recent Advances in Probability and Statistics at [Florida Atlantic University](#) in Boca Raton, FL October 30 – November 1, 2009.
- 2009 ◇ **A New Use of Group Representation Theory in Statistics.** Contributed presenter at the 2009 AMS Sectional Meeting at [Penn State University](#) in University Park, PA October 24 – 25, 2009.
- 2009 ◇ **The First Summer International Workshop on Statistical & Computational Genetics.** Invited short course instructor of this workshop at [Beijing Forestry University](#), Beijing, China, August 2009.
- 2009 ◇ **Constructing Gene Expression Networks.** Invited talk given at [Penn State University](#), Department of Public Health Sciences, Hershey, PA, April 2009.
- 2009 ◇ **Clustering Gene Expressions.** Invited talk presented at the Population Sciences Program Quarterly Meeting at [Penn State Hershey](#), December 2009.
- 2009 ◇ **A Bootstrapped Linearity Test.** Invited talk presented at Food and Resource Economics Seminar at [University of Florida](#), Gainesville, FL, March 2009.
- 2008 ◇ **Linkage Disequilibria Analysis of Multilocus Multiallelic Markers in Outcrossing Populations.** Invited presenter at the World Congress of Gene, [Guangzhou, China](#), December 2008.
- 2008 ◇ **Resampling the Bispectrum.** Invited talk presented at the workshop on Bootstrap and Time Series in [Kaiserslautern, Germany](#), June 2008.
- 2008 ◇ **Nonparametric Function Estimation with Infinite-Order Kernels.** Invited talk presented at SIAM-SEAS Special Session on Recent Developments on Empirical Likelihood and Nonparametric Statistics at the [University of Central Florida](#), March 2008.
- 2008 ◇ **Nonparametric Estimation at a Semiparametric Rate.** Invited presenter at the SRCOS conference on Modern Semiparametric Methods in Action in [Charleston, SC](#), June 2008.
- 2008 ◇ **An Epistatic Model for Mapping Phenotypic Plasticity of a Count Trait.** Talk presented at ENAR, in [Arlington, VA](#), March 2008.
- 2008 ◇ **Nonparametric Estimation at a Nearly Parametric Rate.** Invited talk presented at the UF/FSU Joint Colloquium at [Florida State University](#) in Tallahassee, FL, April 2008.

Presentations (continued)

- 2008 ◇ [Recent Math Problems Arising in Statistics involving Generalized Functions, Group Representations, Inequalities, and the Riemann Zeta Function](#). Invited talk at the Applied Math Seminar in the Department of Mathematics at the [University of Florida](#), April 2008.
- 2008 ◇ [Resources for Statisticians](#). Invited speaker at the Statistics Student Seminar at the [University of Florida](#), Gainesville, FL, November 2008.
- 2007 ◇ [Density Estimation of IID, Dependent, and Censored Data](#). Presentations at [Cal Poly Pomona](#) (Pomona, CA) in March 2007.
- 2007 ◇ [Density Estimation of IID, Dependent, and Censored Data](#). Presentations at [Cal Poly](#) (San Luis Obispo, CA) in March 2007.
- 2007 ◇ [Nonparametric Function Estimation with Infinite-Order Kernels and Applications](#). Presentation at [University of Florida](#) in February 2007.
- 2007 ◇ [Nonparametric Function Estimation with Infinite-Order Kernels and Applications](#). Presentation at [Tulane](#) in February 2007.
- 2007 ◇ [Nonparametric Function Estimation with Infinite-Order Kernels and Applications](#). Presentation at [University of Nebraska, Lincoln](#) in February 2007.
- 2007 ◇ [Nonparametric Function Estimation with Infinite-Order Kernels and Applications](#). Presentation at [University of Nevada, Las Vegas](#) in February 2007.
- 2007 ◇ [A Statistical Journey through Historic Haverford](#). Presentation at [Haverford College](#), Haverford, PA, January 2007.
- 2007 ◇ [Linear Models with R: A demonstration](#). Talk presented for a linear models course at [University of California, San Diego](#) (La Jolla, CA), April 2007.
- 2006 ◇ [Density Estimation of IID, Dependent, and Censored Data via Flat-Top Kernels](#). Talk presented at the Conference on Current Research and Applications of Statistical Methodology – [San Diego](#) Chapter of the ASA, Second Annual One-Day Conference, November 17, 2006.

Other Conferences

Chinese Language & Linguistics

- May 9-10, 2025 ◇ Chinese Language Teachers Association (CLTA) Annual Conference, New York City
- April 5-6, 2025 ◇ 8th International Forum on Linguistics and Chinese Education (IFOLCE) / 32nd International Conference on Chinese Language Instruction at Princeton University
- April 26-27, 2024 ◇ 31st International Conference on Chinese Language Instruction at Princeton University

Statistical & Biomedical

- May 17-20, 2026 ◇ Society for Clinical Trials (SCT) 2026 Annual Meeting, Phoenix, AZ
- May 21-24, 2024 ◇ 8th International Workshop in Sequential Methodologies, Orem, UT
- June 28 – July 1, 2021 ◇ United States Conference on Teaching Statistics: Expanding Opportunities
- January 6-9, 2021 ◇ Joint Mathematics Meetings 2021
- August 3-6, 2016 ◇ MAA MathFest 2016, Columbus, OH
- August 5-8, 2015 ◇ Centennial Celebration MAA MathFest 2015, Washington, DC
- July 13-17, 2015 ◇ "Event History & Survival Analysis" Statistical Horizons workshop in Philadelphia by Dr. Paul Allison
- June 19-20, 2015 ◇ "Longitudinal Data Analysis Using SAS" Statistical Horizons workshop in Boston by Dr. Paul Allison
- June 11-12, 2015 ◇ "Multilevel Modeling: A Second Course" Statistical Horizons workshop in Philadelphia by Dr. Kris Preacher
- May 1-2, 2015 ◇ "Causal Inference with Directed Graphs" Statistical Horizons workshop in Philadelphia by Dr. Felix Elwert
- March 27-28, 2015 ◇ "Multilevel Modeling" Statistical Horizons workshop in Philadelphia by Dr. Tenko Raykov
- November 14-15, 2014 ◇ "Applied Bayesian Data Analysis" Statistical Horizons workshop in Philadelphia by Dr. Shane Jensen

Other Conferences (continued)

- November 11-15, 2013 ◇ Leadership Academy, Hershey, PA
- April 2012 ◇ ENAR 2012 Spring Meeting, Washington, DC
- April 2012 ◇ "Bayesian Adaptive Methods for Clinical Trials" (full-day ENAR workshop), Washington, DC
- March 2012 ◇ Translational Bioinformatics and Clinical Research Informatics Joint Summits, San Francisco, CA
- December 2011 ◇ "First Short Course on Next-Gen Sequencing: Technology and Statistical Methods", Birmingham, AL
- October 2011 ◇ Symposium on Large-Scale Inference, Silver Spring, MD
- July 2010 ◇ First Joint Biostatistics Symposium, Beijing Renmin University, Beijing, China
- July 2010 ◇ An International Conference in Honor of P.L. Hsu's 100th Birthday, Beijing, China
- April 2010 ◇ Frontiers in Mathematical Biology: NSF-NIH PIs Meeting 2010, Washington, DC
- April 2010 ◇ American Association for Cancer Research 101st Annual Meeting, Washington, DC
- February 2010 ◇ Penn State Cancer Institute Mini-Symposium, "Frontiers in the Interface between Statistics and Genetics", Hershey, PA (Co-organizer, Session Chair)
- January 2009 ◇ Winter Workshop on "Semiparametric Methodology", Gainesville, FL
- April 2008 ◇ Princeton Day of Statistics, Princeton, NJ
- April 2008 ◇ K-State Applied Statistics Conference, Manhattan, KS
- January 2008 ◇ Winter Workshop on "Bayesian Model Selection and Objective Methods", Gainesville, FL (Session Chair)
- August 2007 ◇ University of Florida College of Agricultural and Life Sciences Teaching Symposium

Graduate Committees/Directed Studies

PhD Committees

- 2026 ◇ Avantika R. Diwadkar (PSU Bioinformatics and Genomics PhD Student), Committee Member, Statistical Approaches to Infer the Genetic Basis of Cell-Cell Interaction and Phenotype Heterogeneity with Applications in Autoimmune and Neurodegenerative Disease, graduated May 2026.
- 2024 ◇ Shuang Gao (PSU Biostatistics PhD Student), Committee Member, Unraveling Genetic Loci in Addiction and Sex-Differentiated Effects Through Innovative Genomic Analysis, graduated December 2024.
- 2023 ◇ Nour Hawila (PSU Biostatistics PhD Student), Advisor, Confidence intervals for noninferiority testing, Bayesian joint modeling of interrater and intrarater reliability, topics in biostatistical modeling and statistics education, graduated June 2023.
- 2022 ◇ Xingyan (David) Wang (PSU Biostatistics PhD Student), Committee Member, Statistical Methods for Aggregating Trans-Ancestry Genome-Wide Association Summary Statistics and Its Applications to Nicotine and Alcohol Addiction Phenotypes, graduated December 2022.
- 2022 ◇ Chenqi (Stacey) Fu (PSU Biostatistics PhD Student), Committee Member, Bayesian Predictive Hypothesis Testing and Model Averaging, graduated November 2022.
- 2021 ◇ Daniel McGuire (PSU Biostatistics PhD Student), Committee Member, Methods for assessing replicability in GWAS and analyzing large-scale Electronic Health Record datasets, graduated June 2021.
- 2021 ◇ Renan Sauteraud (PSU Biostatistics PhD Student), Committee Member, Statistical Methods for the Functional Genomic Analysis of the X Chromosome, graduated June 2021.
- 2020 ◇ Vishal Midya (PSU Biostatistics PhD Student), Co-advisor with Jason Liao, Frequentist and Bayesian hypothesis testing focusing on meaningful effect sizes, graduated August 2020.
- 2013 ◇ Diana Tancelosky (PSU MD/PHD Program), Committee Member, Applying a Candidate Gene Approach to Investigate the Molecular Changes that Accompany the Development of Addiction-Like Behaviors to Opioids in Rats. Defended September 16, 2013.
- 2013 ◇ Joe Ashmore (PSU Nutritional Sciences), Committee Member, Dietary, Lifestyle, and Genetic Factors Influencing Colorectal Cancer Risk. Defended January 28, 2013.

Graduate Committees/Directed Studies (continued)

- 2013 ◇ Joel Coble (PSU Biomedical Sciences), Committee Member, Distal arthrogryposis type I and MyBPC1, presentation/exam on June 14, 2012, June 28, 2012, and January 24, 2013.
- 2011 ◇ Wei Sun (PSU Statistics), Committee Member, Feature Screening and Variable Selection in Ultra-High Dimensional Survival Data Analysis, advanced December 7, 2011.
- 2009 ◇ Xiaobo Li (UF Forestry and Statistics Co-PhD), Committee Member, Association Mapping for Wood Property Traits of Loblolly Pine using SNP Markers, defended May 18, 2009.
- 2009 ◇ Qin Li (UF Statistics), Committee Member, Statistical Models for Haplotyping Complex Inherited Diseases in Humans, defended July 2, 2009.
- 2009 ◇ Yao Li (UF Statistics), Committee Member, Statistical Designs and Algorithms for Modeling the Genetic Architecture of Cancer Susceptibility, defended July 1, 2009.
- 2009 ◇ Xinxin Zhang (UF Economics), Committee Member, Modeling Yield Loss and Catastrophic Risk in Florida Citrus Industry, advanced June 12, 2009.

Master's Committees

- 2009 ◇ Shuguang Tan (UF Statistics) On L_1 Regularization Path Algorithm for GLMs, defended 6/11/09.
- 2008 ◇ Nasha Wang, (UF Statistics) Committee Chair, Functional Interval Mapping of Quantitative Trait Loci (QTL) in Tumor Cell Growth, defended 12/16/08.
- 2008 ◇ Hong Li, (UF Statistics) Committee Chair, Sequencing Tumor Growth with HapMap, defended 11/25/08.
- 2008 ◇ Guifang Fu, (UF Statistics) Committee Chair, QTL Mapping on Shape, defended 11/24/08.
- 2008 ◇ Anhui Huang (UF Soil & Water) Characterization of Arsenic Resistant Bacterial Communities in the Rhizosphere of an Arsenic Hyperaccumulator *Pteris Vittata* L., defended 10/31/08.
- 2008 ◇ Jiahua Li, (UF Statistics) A Statistical Model for Mapping Quantitative Trait Loci in Autotetraploids, defended 10/20/08.
- 2008 ◇ Liming Qu, (UF Statistics) Tolerance Interval and its application, defended 8/19/08.
- 2008 ◇ Sharon Wen, (UF Statistics) Detection of genetic imprinting effect expressed by multilocus haplotypes, defended 7/28/08.
- 2008 ◇ Hao Zheng, (UF Statistics) Treatments of Ties in the Sign Test, defended 7/22/08.
- 2008 ◇ Jiang Liu, (UF Statistics), Functional Mapping of Quantitative Trait Loci Regulating Pharmacokinetics and Pharmacodynamics by Using Fentanyl Addiction as an Example, defended 7/10/08.

Directed Independent Studies/Internships

- 2019 ◇ Andy Chen (Hershey High School Student), Biostatistical Applications (Summer 2019)
- 2018 ◇ Jay Yang (Penn State Undergraduate), Biostatistical Applications (Summer 2018)
- 2018 ◇ Hui Zeng (Penn State, New Biostatistics PhD Student), Nonparametric Estimation in Survival Analysis (Summer 2018)
- 2013 ◇ Cong Xu (Penn State), Biostatistical Applications (Fall 2013)
- 2013 ◇ Mandi Feinberg (VCU senior), Bioinformatics Applications (Summer 2013)
- 2012 ◇ Allshine Chen (Penn State), Summer Internship (Summer 2012)
- 2010 ◇ Heather Connelly (Ursinus College senior), Earthquake Statistics (Summer 2010)
- 2009 ◇ Che-Shun Chang (UF Statistics Graduate Student), Regression and Multilevel/Hierarchical Models (Summer 2009)
- 2009 ◇ Steve Chung (UF Statistics Graduate Student), Applied Time Series Methods (Spring 2009)
- 2008 ◇ Alex Savenkov (UF Statistics Graduate Student), Time Series Analysis and Applications (Fall 2008)
- 2008 ◇ Yanqing Yan (UF Genetics Graduate Student), Functional Mapping of Quantitative Trait Loci (Fall 2008)
- 2008 ◇ Guifang Fu (UF Mathematics Graduate Student), Statistical Genetics of Quantitative Traits (Spring 2008)
- 2007 ◇ Yingxia Wan (UF Statistics Graduate Student), Subsampling the Bispectrum (Fall 2007)

Courses Taught

Penn State College of Medicine (Penn State Hershey)

- 2013 – 2024 (Spring) ◇ PHS 528: Bayesian Methods
- 2023 – 2024 (Spring) ◇ PHS 583: Asymptotic Tools
- 2011 – 2014 (Spring) ◇ PHS 525: Biostatistics for Lab Scientists

Penn State University (Main Campus)

- 2010 (Spring) ◇ PHS 516/STA 597E: Statistical Genetics (co-taught with Rongling Wu)

Beijing University of Chemical Technology

- 2022 (Summer) ◇ Bayesian Analysis and Statistical Rethinking (short course)
- 2019 (Summer) ◇ Probability, Statistics, and Bayesian Analysis (short course)

MathPath Summer Camp (at Grand Valley State University)

- 2019 (Summer) ◇ An Introduction to R and Monte Carlo Simulation (short course)
- 2019 (Summer) ◇ American Invitational Mathematics Examination Contest Preparation (short course)

University of Florida

- 2009 (Spring), 2008 (Fall) ◇ STA 5325/STA 4321: Introduction to Probability
- 2008 (Spring) ◇ STA 5856/STA 4853: Introduction to Time Series and Forecasting
- 2007 (Fall) ◇ STA 6857: Applied Time Series Analysis

University of California, San Diego

- 2007 (Winter) ◇ MATH 10B: Second Semester Calculus

San Diego Miramar College

- 2006 (Summer) ◇ MATH 252: Calculus with Analytic Geometry III
- 2006 (Spring) ◇ MATH 96: Intermediate Algebra and Geometry
- 2005 (Fall) ◇ MATH 119: Elementary Statistics

Awards

- 2014 ◇ Dean's Award for Excellence in Teaching, Penn State College of Medicine
- 2014 ◇ Innovation in Team Science Award, Penn State Faculty Research Recognition Awards
- 2013 ◇ Society of Distinguished Educators Junior Faculty Teaching Award (Finalist)
- 2008 ◇ Research Innovation Award, University of Florida Institute of Food and Agricultural Sciences (IFAS)
- 2005 – 2006 ◇ Outstanding TA Award, UCSD Mathematics Department, for “exceptional contributions to the department’s instructional efforts”

Service

- 2026 – 2030 ◇ Elected Faculty Senator, Penn State Faculty Senate
- 2024 – 2026 ◇ Member, Penn State University Promotion and Tenure Review Committee
- 2021 – ◇ Faculty Development Advisory Committee Member
- 2021 ◇ Special Member for National Eye Institute Clinical Review Meeting; reviewer for three grants during their March 2021 study section
- 2021 – 2023 ◇ Advisory Board Member for *Academic Journal of Applied Mathematical Sciences*
- 2020 – 2024 ◇ Member, Scientific Merit Review Panel for Healthcare Informatics Health Services Research at the Department of Veterans Affairs; review healthcare informatics grants twice annually
- 2020 – ◇ Editorial Board Member for *Open Journal of Statistics*

Service (continued)

- 2012 – ◇ Member, Public Health Sciences Program Directors & Executive Committee
- 2019 – 2020 ◇ Reviewer, Scientific Merit Review Panel for Healthcare Informatics Health Services Research at the Department of Veterans Affairs
- 2015 – 2019 ◇ Member, Penn State Faculty Senate
- 2010 – 2019 ◇ Member, Penn State Cancer Institute Scientific Review Committee
- 2016 ◇ Reviewer, NIH Study Section: Biomedical Computing and Health Informatics (BCHI)
- 2013 – 2016 ◇ Member, IT Research Computing & Informatics Advisory Council
- 2013 – 2016 ◇ Member, Curriculum Education Committee (medical education)
- 2012 – 2016 ◇ Member, Penn State’s Global Engagement Network Committee to Partnership with Peking University
- 2010 – 2016 ◇ Member, College of Medicine Scientific Review Committee
- 2010 – 2016 ◇ Member, Institutional Review Board (Board A)
- 2013 – 2014 ◇ Member, Biomedical Sciences PhD Program Core Curriculum Revision Committee
- 2012 – 2013 ◇ Member, Data Management Symposium Planning Committee
- 2012 – 2013 ◇ Bioinformatics Search Committee at Penn State
- 2012 – 2013 ◇ Guest Editor, Computational and Mathematical Methods in Medicine
- 2012 ◇ IT Manager III Search Committee
- 2011 – 2012 ◇ Bioinformatics Search Committee at Penn State
- 2011 – 2012 ◇ Member, Research Use of Clinical Data Committee
- 2009 – 2012 ◇ Member, Research Computing Advisory Committee
- 2010 – 2011 ◇ Biostatistics Search Committee at Penn State
- 2010 ◇ Member, Research Informatics Advisory Group
- 2010 ◇ Member, Penn State College of Medicine Research Informatics Advisory Group
- 2010 ◇ Co-Organizer of the mini-symposium on “Frontiers in the Interface between Statistics and Genetics” at Penn State University, February 25, 2010.
- 2007 – 2009 ◇ Seminar Chair, Department of Statistics, University of Florida